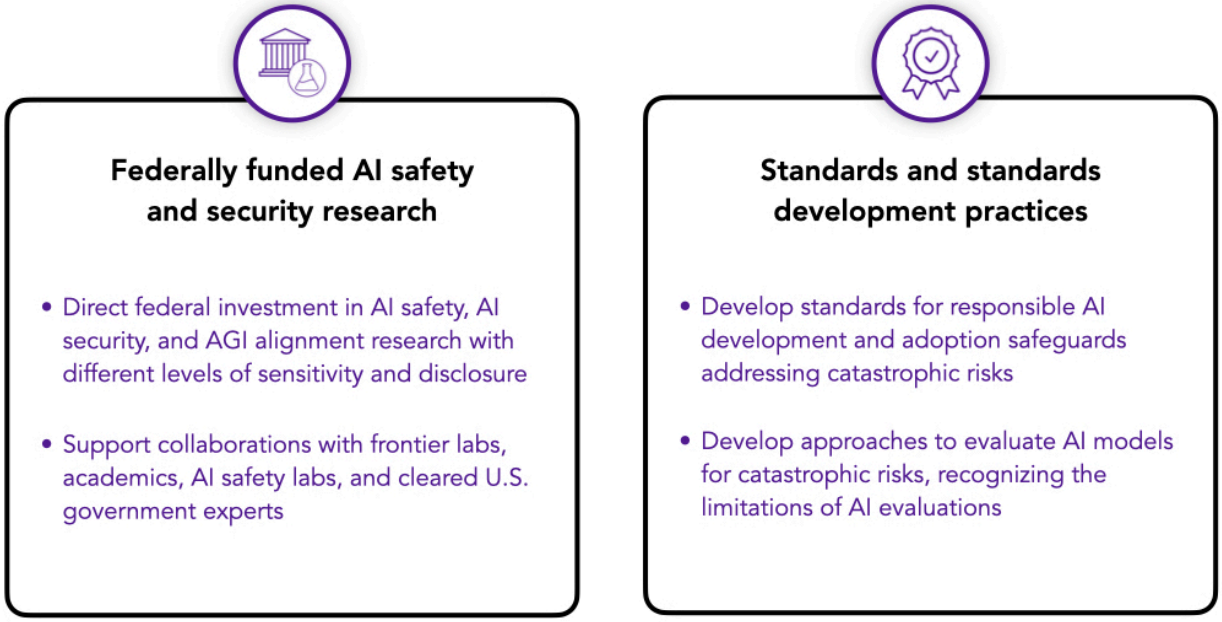
LOE3: Increase national investment in technical AI safety research and standards development



The acceleration of investment in AI capabilities is outpacing the development of proportionate technical safeguards against advanced AI and AGI risks [5] (Introduction, 0.5.1.2 and 0.5.1.3). If this continues, frontier AI labs may find themselves unable to meet the safety and security challenges posed by their own systems (Introduction, 0.5.1.4). Unless strong technical safeguards are designed, standardized, and broadly applied, continued development and adoption of frontier AI systems could create significant risks (Introduction, 0.5.1.1).

This LOE outlines specific actions the U.S. government could take to **strengthen domestic technical capacity in advanced AI safety and security, AGI alignment, and other technical AI safeguards.** These actions include:

- Directly funding **advanced AI safety and security research** including AGI-scalable alignment research (see 3.1); and

- Developing, regularly reviewing, and promulgating **safety and security standards** for responsible AI development and adoption (see 3.2).

These solutions would grow the pool of competent AI and AGI safety and security researchers, encourage organizations to invest in AI safety and security, and increase overall understanding of AGI alignment techniques.

3.1 Federally funded research in AI safety and security and AGI-scalable alignment

Since 2021, annual private spending on AI capabilities has exceeded $100B, dramatically outstripping funding for AI safety and security [184]. AI capabilities research publications outnumber safety publications by a fifty-to-one margin [184]. Yet according to experts, in order for the level of investment in AGI-scalable safety research to approach the level needed to address catastrophic AI risk, it would need to reach at least 50% of total research spend [185]. The current funding gap exists because, while AI safety and security is a common good, AI capabilities produce benefits that are easier to privatize. As a result, private industry has a strong incentive to invest in capabilities at the expense of research aimed at catastrophic AI risk reduction (Introduction, 0.5.1.4, 0.5.1.5, and 0.5.3.1).

Aside from private industry and academic labs [186,187], there is also an emerging ecosystem of independent AI safety researchers, nonprofit and for-profit AGI alignment labs [188–190], and advanced AI auditing firms [94,166,191]. However most of the organizations in this ecosystem currently rely on a small set of donors for funding, which could limit the diversity of strategies to AGI alignment reflected in the output of the AI safety research community.[48] Because AGI alignment is still a nascent field that benefits disproportionately from a breadth of research approaches, this concentration of funding sources could create limitations to progress.

The U.S. government is uniquely positioned to close the funding gap between AI capabilities and AI safety and security, accelerate progress on core problems, and diversify the set of funding sources and research programs aimed at solving AGI alignment.[49] See Annex L: AI safety and security research topics for a list of research areas that could be supported by U.S. federal funding.

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[48] See Annex E: Funding in AI safety for more information.

[49] See Annex B: The full challenge of AGI alignment for details.

3.1.1 Levels of AI safety and security research

In addition to its capacity for funding critical basic research, the U.S. government can also ensure that those research efforts adhere to necessary security practices and that they have the required levels of access to key technologies, resources, and talent. This is critical in the context of AI safety and security research that may require interacting closely with potentially dangerous frontier AI models.

Not all such research will require the deepest levels of access. We recommend establishing tiers of AI safety and security research that reflect the escalating risk and responsibility associated with granting researchers broader access to increasingly capable models and evaluation capabilities. We provide a possible categorization below along with examples.

3.1.1.1 Open-source and public research

Some types of AI safety and security research involve only low-risk activities and do not require any special access to frontier AI models. All aspects of research at this level can and should be publicly shared. Funding of open-source research could support independent AI alignment and AI safety and security researchers and organizations. It could also support the development of software tools and frameworks with important safety and security applications, such as test harnesses, evaluation frameworks, sandboxes, and interpretability tools. To maximize the diversity of research approaches, we recommend that funding grants at this level impose minimal compliance overhead on recipients.

Some examples of research at this level could include:

- Certain kinds of evaluations for advanced AI systems, including basic sandbox and testbed development;
- Certain kinds of AI interpretability research, including on sub-frontier AI models generally regarded as safe [192];
- Theoretical research aimed at investigating evidence for risks associated with power-seeking and associated tendencies [2,47];
- Research aimed at determining the extent to which existing alignment techniques could or could not scale to AGI-level systems [84]; and

- Basic research into on-device monitoring for AI chips [148].

The decision to classify AI safety research as open-source should take into account the risk that the research could empower threat actors to weaponize AI more effectively, subvert on-chip monitoring capabilities (e.g. LOE1, 1.5.3), or otherwise accelerate or proliferate AGI development. But it should also account for the potential economic and societal harms of withholding such research from the public (Introduction, 0.4.2). As a result, publication decisions should be closely informed by input from academic researchers, the AI safety and national security communities, and private sector industry including the startup community.

Funding for open-source and public research could be directed towards the following types of entities:

- Academic labs and independent researchers working on AI safety and security projects;
- Nonprofits conducting AI safety and security research;
- Researchers at National Labs and other centers of excellence within the U.S. government with the capacity to pursue AGI-scalable alignment research;
- For-profit AI safety research labs working on AGI-scalable alignment techniques.

3.1.1.2 Research that requires access to proprietary AI models

Certain essential types of AI safety and security research will require a high degree of access to frontier labs' proprietary AI models, or may otherwise involve moderate-risk activities without being sensitive enough to necessitate a security clearance. Some of the results of research at this level could still be published, but the extent of disclosure should be informed by risk assessments and IP considerations.

The government can play a key role in supporting this research by offering direct funding, mediating researcher access to developers' AI models, and overseeing researcher compliance with the necessary security practices. This could involve certifying AI safety research groups or individual researchers to access frontier labs' proprietary AI models. This approach could also be used to support AI evaluator certification and access to frontier lab models by entities such as the ASTF (see LOE1, 1.4.1.2) or a permanent regulator (see LOE4, 4.1.2.1).

Some examples of research at this level could include:

- Research that involves inducing powerful AI models to behave in dangerous ways, such as behavioral and propensity evaluations for dangerous capabilities like deception and persuasion [193];

- The development of some sets of private AI evaluations that should be withheld from frontier labs to avoid compromising the integrity of model audits (see 3.2.2);

- Some kinds of fundamental AGI alignment research, particularly if they require direct access to frontier models;

- Mechanistic interpretability research that requires direct access to frontier models [194]; and

- Research into on-device monitoring for AI chips that could benefit from frontier labs' engineering resources or expertise in scaled AI training runs [148].

Given the potential sensitivity of research in this category, researchers and model evaluators may require dedicated infrastructure to support secure access to the weights of frontier AI models. We describe a high-level weight-sharing protocol that could meet this requirement in Annex M: Secure temporary storage of model weights. In addition to directly funding private collaborations with AI labs, the government could fund the development of secure information-sharing infrastructure to ensure this research can occur securely and with minimal friction.

3.1.1.3 National security research

The most high-risk category of AGI safety research will likely involve unrestricted access to frontier AI models to evaluate propensities for highly dangerous capabilities, including WMD-like and WMD-enabling capabilities. This category of research will need to be conducted in a classified environment by cleared researchers following appropriate security procedures. Some kinds of model evaluations overseen by entities such as the ASTF (see LOE1, 1.4.1.2) or a permanent regulator (see LOE4, 4.1.2.1) may need to be classified at this level.

Similarly to proprietary research (see 3.1.1.2), this level of research would enable a capability for secure sharing of the weights of frontier AI models. See Annex M: Secure

temporary storage of model weights for a description of a high-level protocol for weight-sharing.

3.1.2 Organizational framework

The U.S. government would need the ability to identify, fund, and administer promising research projects at each of the sensitivity levels in 3.1.1. Each level may have a different set of oversight, administration, and funding conditions that reflect its degree of risk and access requirements.

The U.S. government has already begun developing a capacity that could support such activities. In January 2024, the National Science Foundation (NSF) launched a pilot program for a National AI Research Resource (NAIRR) [195,196], whose budget is estimated to reach $2.6 billion over an initial six-year period. The NAIRR also includes an initiative called NAIRR Secure, which aims to support research involving sensitive data by assembling secure compute clusters and other privacy-preserving resources [197]. One of the NAIRR's four key goals is to **advance trustworthy AI** [198].

Many of the AI safety and security research areas relevant to mitigating catastrophic risk from advanced AI are aligned with this NAIRR key goal. As a result, we believe that relevant research directions at each of the relevant sensitivity levels (see 3.1.1) could be directly supported by:[50]

- The NAIRR, through a dedicated National Center for AI Alignment and Security Research (NCAASR) established under the NAIRR;
- A new AI Alignment and Security (AAS) Federally Funded Research and Development Center (FFRDC), possibly in collaboration with the NAIRR;[51] or

[50] Given that there are ongoing U.S. government workstreams in this area, we recommend considering these options in conjunction with other entities that could already be in development.

[51] One advantage of an AAS FFRDC is that it could be sponsored by organizations such as the Defense Advanced Research Projects Agency (DARPA), the Intelligence Advanced Research Projects Activity (IARPA), or the DOE's National Laboratories, which all have established CBRN research capabilities that could support in designing AI evaluations for frontier models. An AAS FFRDC would also offer the government a classified and secure environment in which researchers can be paid industry-competitive salaries. This is a particularly important factor given the prevailing level of compensation at frontier AI labs (see LOE1, 1.4.2).

- Both a NCAASR and a new AAS FFRDC, with research areas divided between them and research collaborations in areas of mutual interest.

We recommend that these Centers establish a review process to determine which kinds of safety research should remain unpublished, and base decisions on national security risk assessments (LOE4, 4.1.4).[52] This review process could also be informed by an ASTF's work under LOE1, 1.4.1.3 and be developed in consultation with members of the technical AI safety community.

These Centers could also advance basic scientific research into AGI alignment, normalize the field, and grow the overall pool of researchers.

3.1.2.1 Considerations for AGI alignment research

Many alignment techniques that work on less capable AI systems are not expected to work on AGI-level systems [84] (Introduction, 0.5.1.4). It is also easier to make measurable progress on sub-AGI alignment than on AGI-scalable alignment. As a result, we recommend that these Centers define and fund research workstreams fully and exclusively dedicated to developing alignment techniques that are intended to scale to AGI-level systems. Absent a clear institutional prioritization of AGI-scalable alignment, research effort funded by the Centers risks being expended in areas in which progress is more legible but less scalable.[53]

See Annex B: The full challenge of AGI alignment for a discussion of the reason why alignment techniques that work on advanced AI systems may fail once these systems reach high enough capability levels. See Annex L: AI safety and security research topics for a list of possible research topics that an NCAASR and AAS FFRDC could fund or support.

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[52] This could also be done in coordination with the I&W and contingency planning efforts in LOE2, 2.3 and 2.4.

[53] Moreover, many forms of sub-AGI alignment research deliver immediate economic returns by making it easier for users to extract value from better-aligned, sub-AGI scale systems. As a result, these research areas already benefit from substantial investment by private-sector frontier AI labs.

3.1.2.2 Research collaborations

Centers such as the NCAASR and/or AAS FFRDC could be instrumental in developing a regime of technical RADA safeguards (LOE1, 1.3.2) to mitigate potential catastrophic AI risks from weaponization and loss of control. The Centers could collaborate with the NIST U.S. AISI to standardize this safeguards regime (see 3.2.3), and with entities such as an ASTF (LOE1, 1.4.1.2) or future regulatory body (LOE4, 4.1.2.1, 4.1.2.2, and 4.1.2.4) to implement and oversee those standards. These Centers could also collaborate with and support AI safety and security initiatives across the U.S. government and beyond, including:

- The BIS in periodic reviews of export controls related to catastrophic AI risk mitigation, including support with on-chip monitoring initiatives (LOE1, 1.5.3);
- The Department of Commerce and DHS for export license red teaming exercises (LOE1, 1.5.3);
- The IC to support creating and updating I&W frameworks (LOE2, 2.3.3);
- The NSC to support creating and updating contingency plans (LOE2, 2.4);
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- The ASTF to help prepare for a future regulatory regime (Annex K, K.2); and
- Through the Department of State, international partners such as the U.K. Frontier AI Taskforce and the U.K. AI Safety Institute (U.K. AISI) [199].

Operationally, with the support of the NAIRR, the Centers could also foster direct collaboration between independent AI safety and security researchers, academics, and industry, including frontier labs. This could include funding the development of infrastructure to support secure information sharing with frontier labs (e.g., NAIRR Secure [197]). It could also offer researchers access to computing resources [198].

3.1.2.3 Sensitive research areas

Certain types of research aimed at mitigating catastrophic weaponization risk from CBRN, cyber, and other applications enabled by advanced AI will require direct access to frontier models, secure information environments, and cleared personnel (see 3.1.1.3). This may also be the case for some categories of research investigating the risk of loss of control due to alignment failure of high-capability AI systems.

The level of security required to support such research will likely require close coordination between frontier AI labs, the teams conducting the research, the IC, and the Centers supporting the research. The Centers would also need infrastructure to support highly secure sharing of information with frontier labs, including model weights (see 3.1.1.3).

Given the substantial overlap in requirements and likely collaboration in other areas, the Centers could also consider sharing infrastructure between sensitive AI research initiatives and the regulatory bodies overseeing private-sector AI scaling, including model evaluations for CBRN and other capabilities (Annex K, K.1 and LOE4, 4.1.3.4.3).

3.2 Standards for AI evaluations and RADA safeguards

Without clear standards for responsible AI development and adoption (RADA), advanced AI developers lack principles against which to judge the adequacy of their safety and security practices. Self-regulatory industry bodies such as the FMF (Introduction, 0.4.1) could have a positive influence on the development of such standards. But competitive pressures — including from challenger labs whose models lag the public frontier by 12 months or less, and are even less rigorous in their safety measures [51,200,201] (Annex D, D.1) — ultimately drive even the frontier labs to employ inconsistent and inadequate safety and security practices (Introduction, 0.5.1.4, 0.5.1.5, and 0.5.3.1).

It is therefore crucial to develop standards for RADA safeguards which address catastrophic risks from weaponization and loss of control in a rigorous and technically informed manner. These should include standards for the evaluation of AI models' behaviors and propensities to display dangerous capabilities [46]. NIST, through the U.S. AISI [202], is currently spearheading U.S. government efforts to develop and promulgate AI evaluation standards consistent with the NIST AI Risk Management Framework [203].

In addition to AI evaluations, standards for effective RADA safeguards will also need to comprehensively address security, risk governance, incident reporting, and other key issues across multiple layers of the advanced AI supply chain. See LOE1, 1.3.2 for our recommendations on guiding principles behind these standards.

Finally, ongoing refinement and updating of standards for RADA safeguards could be supported by expanded national research capacity in AI safety and security research as described in 3.1.

3.2.1 Limitations of AI evaluations

AI evaluations are important tools to increase confidence in the safety and security of advanced AI systems and support RADA safeguards. And depending on the degree of coverage and prior real-world experience with an evaluation set, passing a set of AI safety evaluations may offer a strong positive indicator of the safety of an AI system. But AI evaluations are inadequate on their own to fully assure that an AI system or model is safe to develop or deploy, because they suffer from a number of fundamental limitations in coverage and reliability [204]. We list below some of these key limitations, and discuss ways to mitigate their impacts in 3.2.2.

3.2.1.1 AI evaluations are not comprehensive

Current AI evaluations **cannot provide comprehensive coverage** of an advanced AI model's behavior or risk surface. If there is a dangerous behavior or set of behaviors that is not included in the test suite used to evaluate a model, then that behavior will not be detected, and it could be displayed or elicited under real world conditions [206] including under high-risk conditions.

There is also currently no way to reliably predict the emergence of capabilities, dangerous or otherwise, from advanced AI systems as they are scaled or augmented [165,206]. This means that not only do researchers have no way of knowing in advance which kinds of evaluations will be most informative for future AI systems, but they also have no way to reliably predict how an *existing* AI system will behave if it is fine-tuned or otherwise augmented. In particular, open-access AI systems can be augmented in any number of ways that may be entirely unanticipated by the system's original developers or the evaluators themselves (Introduction, 0.5.1.6). Therefore, as limited as AI evaluations are for substantiating the safety of proprietary AI systems, they are even more limited when applied to open-access AI systems.

3.2.1.2 AI evaluations cannot confirm that a dangerous capability is absent

AI evaluation techniques today are empirical. They generally function by feeding a model a wide range of input prompts and observing its outputs.[54]

[54] More sophisticated techniques may fine-tune a model to attempt to elicit latent capabilities. This is a more effective approach than prompting, but is still limited.

If the model returns a dangerous output in response to an evaluation, this may show that the model has a dangerous capability. But if the model does not return a dangerous output in response to the evaluation, this could be because the evaluator failed to identify an effective prompt, or for other reasons, rather than because the model does not have the evaluated capability. Even small changes to prompting techniques often reveal latent capabilities that were not detected by prior evaluations [207]. As a result, AI evaluations **can only reveal the presence, but not confirm the absence,** of dangerous capabilities [204].

3.2.1.3 AI evaluations are highly vulnerable to manipulation

AI evaluations **can be undermined and manipulated easily**. If an AI model fails an evaluation, its developer can fine-tune the model — or make other superficial changes to it — until the model passes that evaluation, without addressing the underlying cause of the failure.

For example, suppose an AI model fails a cyberattack evaluation by complying with a user request to write working malware. In this case, it is easier for the model's developer to train it to refuse the particular user request it was tested on, than to rigorously investigate and address the underlying cause of the behavior. But this superficial approach could leave the model vulnerable to jailbreaks [83,208] that induce the model to display the undesirable behavior under slightly more complex conditions. In other words, the dangerous capability can remain latent, even if an AI model passes the evaluation. Moreover, if an AI developer is able to resubmit fine-tuned or "patched" models for evaluation an arbitrary number of times, the resubmitted models may eventually pass the original evaluations purely or partly by chance [46].[55]

Given the prevailing competitive environment, frontier AI labs and other advanced AI developers face a strong incentive to game evaluations in this way (see 0.5.3.1). In fact, by the private judgment of one expert with direct knowledge of the matter, this has already begun to occur at one major frontier AI lab. Moreover, AI developers also face related incentives to argue that models that pass unreliable evaluations should still be considered safe for the purpose of further development or deployment.

[55] One frontier AI researcher stated during a private discussion that passing a dangerous capability evaluation provides little evidence that the model is safe, if the lab developing the model is allowed to resubmit tweaked versions of the model for re-evaluation until the evaluation is passed. As they put it, "There's a key step [missing] here of figuring out what went wrong."

3.2.1.4 AI evaluations could fail systematically in high-capability regimes

At higher levels of capability, advanced AI and near-AGI systems could develop the capability to infer whether a given interaction is part of their training, their testing, or their deployment. This property is called **situational awareness** [209,210], and researchers at major frontier labs believe it could arise as an emergent capability in future AI systems [209].

If an AI system can infer that it is being tested, it may have the ability to produce different outputs when it is being tested than when it is deployed. In principle this could allow the AI model to **manipulate its own evaluations**, by producing apparently safe outputs in response to evaluation queries in testing, and unsafe outputs in response to real user queries in deployment [209]. This hypothesized behavior is known as **deceptive alignment** [45] . Deceptive alignment remains a speculative and controversial risk [211], but if it manifests, it could systematically undermine the entire effectiveness of a naive AI evaluations regime.[56]

3.2.2 Addressing the limitations of AI evaluations

There are several possible strategies to lessen the impacts of the above limitations on the value of AI evaluations as safety signals.

First, a diverse ecosystem of independent AI model evaluators is crucial. **Independence** is crucial because model developers are heavily incentivized to pass safety evaluations by making superficial changes that do not address underlying drivers of risk (see 3.1.2.3). Independent evaluators can administer **private evaluation sets**, whose exact protocols are not known to AI model developers, and that are therefore more difficult to pass by superficial means. Independence also means that AI evaluators **should not be selected by the AI developers** they are evaluating, to avoid obvious conflicts of interest. **Diversity**, on the other hand, is crucial because current AI behavioral evaluations are unreliable, and failing to observe a dangerous capability may not mean that the capability is actually absent (see 3.2.1.1). A diverse ecosystem of evaluators supports **multiple independent evaluation strategies** and this combination is more likely to detect a dangerous capability if one is present.

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[56] It is also important not to anthropomorphize these risks. Deceptive alignment is not expected to arise due to sentience, human-projected drives, or consciousness, but rather because, under certain circumstances, it may simply be an effective behavior for the purpose of achieving the goals an AI system has internalized during its training.

Second, a diverse set of AI evaluations, with **different private subsets** developed, refined, and administered by different evaluators, is also crucial. This is because there is no way to fully assess the capability surfaces of modern advanced AI models (see 3.2.1.1). The broader the set of teams, and the more diverse the evaluations, the lower the chance that dangerous capabilities and other risk vectors are unexamined. This also means that standards bodies should avoid overspecifying the parameters of AI evaluations — with the exception of the security conditions under which they should be administered.[57]

Finally, because AI evaluations may have fundamental limitations (see 3.2.1.4), over-reliance on AI evaluations could propagate a false sense of security among AI developers, regulators. Instead, AI evaluations could be considered as **one part of a multifaceted case** for model safety and security, consistent with the principle of defense in depth.

3.2.3 Standards for RADA safeguards under catastrophic risk

For recommended principles to guide standards development for RADA safeguards under catastrophic risk, see LOE1, 1.3.2.

For an example of a complete standards framework for RADA safeguards, including sample calculations of thresholds for covered entities, see LOE4, 4.1.3.

[57] Because dangerous capability evaluations often involve actively eliciting dangerous capabilities, in high-capability regimes the evaluations themselves could become sources of risk and should therefore observe strict operational security precautions.